

Experiment 19: Edge Detection using Sobel Matrix along XY-axis

AIM

To detect edges using Sobel operators in both X and Y directions.

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Apply Sobel X and Sobel Y.
4. Combine both results.
5. Display the output.

PROGRAM

```
import cv2

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobelxy = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

cv2.imshow("Original Image", img)

cv2.imshow("Sobel XY", sobelxy)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

OUTPUT



Explanation

We detected edges along both the X and Y directions using the Sobel operator. The two outputs were combined to highlight edges in all directions.